Translating Verbal Phrases into Algebraic Expressions

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To **translate verbal phrases into algebraic expressions**, identify the operation described by the words and represent the unknown quantity with a variable.

Two Steps

Step 1: Find the operation hidden in the words.

Step 2: Use a variable for the unknown number.

"a number" $\rightarrow x$

Common Words and Phrases

Certain words tell us which operation to use.

- Addition
- Subtraction
- Multiplication
- Division

Addition

sum, plus, added to, increased by, more than

$$x + 5$$

Subtraction

difference, minus, decreased by, less than, subtracted from

$$x - 5$$

Multiplication

product, times, multiplied by, twice, thrice, double, triple

$$5x$$

Multiplication Words

Some words already tell us the number.

twice, double	$2x$
thrice, triple	$3x$

Division

quotient, divided by, the ratio of, half of

$$\frac{x}{5}$$

Half Of

Half of a number means divide the number by 2.

$$\text{half of a number} = \frac{x}{2}$$

Summary Table

Operation	Symbol
Addition	$+$
Subtraction	$-$
Multiplication	$\times$
Division	$\div$

Watch Out: Less Than

Less than reverses the order.

"5 less than a number"

$$x - 5$$

The number comes **first**.

Watch Out: Difference Between

Difference between keeps the order of the words.

"the difference between a number and 4" $\rightarrow \ x - 4$

Watch Out: Quotient

Quotient of keeps the order of the words.

"the quotient of a number and 4"

$$\frac{x}{4}$$

The first number is on top.

Watch Out: Sum Inside Another Operation

When a sum or difference is multiplied or divided, use **parentheses**.

"3 times the sum of a number and 2" $\rightarrow \ 3(x+2)$

Remember: Twice Then Add

Read the phrase from the inside out.

"3 more than twice a number"

twice a number $= 2x$

3 more than that $= 2x + 3$

Examples

Example 1

The sum of a number and 5.

$$x + 5$$

Sum means add.

Example 2

A number decreased by 4.

$$x - 4$$

Decreased by means subtract.

Example 3

Six times a number.

$$6x$$

Times means multiply.

Example 4

The quotient of a number and 3.

$$\frac{x}{3}$$

Quotient means divide.

Example 5

Seven less than a number.

$$x - 7$$

Less than reverses the order.

Example 6

Three more than twice a number.

$$2x + 3$$

Twice a number is $2x$.

Then add 3.

Example 7

Half of the sum of a number and 8.

$$\frac{x+8}{2}$$

First find the sum $x+8$.

Then take half of it.

Practice Exercises

Practice 1

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

The sum of a number and 12.

Practice 2

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

A number decreased by 8.

Practice 3

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

Five times a number.

Practice 4

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

The quotient of a number and 7.

Practice 5

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

Nine more than twice a number.

Practice 6

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

Six less than three times a number.

Practice 7

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

The difference between a number and 15.

Practice 8

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

Half of the sum of a number and 10.

Practice 9

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

Four times the difference between a number and 3.

Practice 10

Translate each verbal phrase into an algebraic expression. Use x to represent the unknown number unless otherwise indicated.

The quotient of the sum of a number and 14 and 2.

Activities

Activity 1

Translate each verbal phrase into an algebraic expression.

The sum of a number and 17.

Activity 2

Translate each verbal phrase into an algebraic expression.

A number decreased by 11.

Activity 3

Translate each verbal phrase into an algebraic expression.

Eight times a number.

Activity 4

Translate each verbal phrase into an algebraic expression.

The quotient of a number and 9.

Activity 5

Translate each verbal phrase into an algebraic expression.

Thirteen more than twice a number.

Activity 6

Translate each verbal phrase into an algebraic expression.

Seven less than four times a number.

Activity 7

Translate each verbal phrase into an algebraic expression.

The difference between a number and 19.

Activity 8

Translate each verbal phrase into an algebraic expression.

One-third of the sum of a number and 12.

Activity 9

Translate each verbal phrase into an algebraic expression.

Five times the difference between a number and 6.

Activity 10

Translate each verbal phrase into an algebraic expression.

The quotient of the sum of a number and 18 and 3.

Thank you!